

Erdős Problem #322

EP-322

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $k \geq 3$ and $A \subset \mathbb{N}$ be the set of k th powers. What is the order of growth of $1_A^{(k)}(n)$, i.e. the number of representations of n as the sum of k many k th powers? Does there exist some $c > 0$ and infinitely many n such that $1_A^{(k)}(n) > n^c$?

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>